

Cole Cappello

colecappello12@gmail.com (201) 874-3632 Bozeman, MT

Education

M.S. in Statistics

Montana State University

GPA: 3.95/4.00

Bozeman, MT

August 2024 - Present

Bachelor of Arts in Mathematics and Philosophy

Rutgers University

GPA: 3.80/4.00

New Brunswick, NJ

September 2018 - May 2022

Awards: Magna Cum Laude, Deans List, Outstanding Graduate Student

Course Work: Linear Algebra, Differential Equations, Multivariable Calculus, Introduction to Computer Science, Probability, Real Analysis I & II, Mathematical Statistics, Abstract Algebra, Linear Models, Advanced Regression Analysis, Stochastic Processes, Bayesian Data Analysis, Sampling

Programming Languages: R, Stan, Python, M, Java

Research Experience

A State-Space Approach to Modeling Tire Degradation in Formula 1

Independent Masters Research Project

Bozeman, MT

January 2025 - Present

- Developed a novel Bayesian state-space modeling framework using Stan to estimate latent tire degradation dynamics in Formula 1 racing from publicly available timing data
- Implemented Hamiltonian Monte Carlo for parameter estimation and posterior inference on time-varying degradation processes
- Designed rolling-origin cross-validation scheme and evaluated predictive performance using RMSPE and Continuous Rank Probability Score
- Developed extensions to a base model and demonstrated improved predictive performance relative to an AR(1) benchmark

Statistical Consulting Seminar

Consulting Project

Bozeman, MT

September 2025 - Present

- Met with a domain expert to determine the proper statistical methodology for answering their research questions
- Implemented a cumulative logistic regression model to understand the effect of various sociodemographic covariates on the probability of having a certain number of risk factors for Metabolic Syndrome among Latino immigrants in Montana

Rutgers Department of Agricultural Economics

Research Assistant

New Brunswick, NJ

October 2021 - August 2022

- Aggregated US county and state-level crop production data and NOAA climate data from 1997 to 2017 using the python pandas library and the USDA NASS Quick Stats API
- Performed regression analyses with R to understand relationships between crop production and climate data

Publications & Manuscripts

- Cappello, C. and Hoegh, A. (2026). "A State-Space Approach to Modeling Tire Degradation in Formula 1 Racing." *Journal of Sports Analytics*. doi:10.1177/22150218261446170.

Work Experience

Montana State University

Graduate Teaching Assistant

Bozeman, MT

August 2024 - Present

- Effectively communicate statistical ideas to undergraduate students using R to help visualize important concepts
- Develop and manage meaningful relationships with students and other instructors
- Timely completion of grading responsibilities and creation of student assessments

Montana Conservation Corps
Americorps Volunteer Forestry Crew Member

Kalispell, MT
May 2023 - November 2023

- Collected data on tree quantity, species, growth, etc. to evaluate forest health while working closely with Kootenai National Forest professionals
- Completed fuels reduction projects, e.g. using chainsaws to perform slash cuts and thin overgrown sections of forest
- A typical 8-day work period involved camping in remote backcountry areas and strenuous hiking for 10 hours each day

Epic Systems
Solutions Engineer

Madison, WI
September 2022 - May 2023

- Wrote M code to analyze historical data on patient treatment procedures and determine average allowed amounts for a customer's self-service pricing tool
- Added functionality to company software to handle the insurance claim transfer process for a new category of clients
- Managed relationships and expectations of a client with respect to deadlines and implementation possibilities

Extracurricular Activities

Alpinism

- Summer 2024 - Climbed The Regular Route of Fairview Dome in Yosemite National Park, one of the 50 classic North American rock climbing routes
- Summer 2023 - Climbed The Lion King, a 5.9/5.10 technical alpine route ascending 1500 feet of slabs and cracks up a large granite face in the panhandle of Idaho
- Summer 2021 - Traveled to Colorado where I summited Mt. Blanca, a 14,000 ft peak featuring exposed ridgelines and class 3 climbing

Skills, Activities, & Interests

Certifications: Basic First Aid & CPR, S-212 Wildland Firefighting Chainsaw

Activities & Interests: Rock Climbing, Skiing, Backpacking, Classic Literature, Short Story Writing, Woodworking